DANCAN DOMINIC

Identity & Access Management (IAM) Engineer

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Professional Profile

Identity and Access Management engineer focused on cloud and hybrid identity security, authentication, access governance, and identity lifecycle automation. Brings three years of cybersecurity and security-tooling experience, combining offensive-security thinking with practical engineering to reduce excessive access, improve auditability, and strengthen identity controls. Hands-on portfolio work spans Microsoft Entra ID, AWS IAM, Keycloak, LDAP-based directories, SSO federation, policy analysis, and automated provisioning workflows.

Professional Experience

Cybersecurity Research & Security Tooling, *Independent / Portfolio*

- Develop and maintain open-source security tooling for reconnaissance, vulnerability assessment, system analysis, and repeatable security workflows.
- Apply Python and Bash automation to reduce manual effort, standardize execution, and improve operational visibility across security tasks.
- Use an attacker-informed approach to assess access paths, identify weaknesses, and translate findings into practical hardening and defensive recommendations.
- Build and document IAM-focused engineering projects covering identity lifecycle automation, federation, access-policy analysis, directory synchronization, and identity threat detection.
- Use version control and structured documentation to support traceability, reproducibility, and iterative improvement of technical work.

Education

BSc Computer Science, *Chuka University* 2026

Bachelor of Science in Computer Science at Chuka University, Kenya, with academic grounding in software development, data structures, networking, and operating systems. Integrates these fundamentals with hands-on cybersecurity work to develop secure, automated, and scalable approaches to identity management and infrastructure protection.

Core IAM Competencies

Identity platforms
Microsoft Entra ID, AWS IAM, Keycloak, OpenLDAP, Samba, Active Directory concepts

Governance & authorization
RBAC, ABAC, least privilege, access reviews, permission boundaries, entitlement auditing

Identity protection
MFA, Conditional Access, session/token controls, identity threat detection, incident response

Authentication & federation
OAuth 2.0, OpenID Connect (OIDC), SAML 2.0, JWT, PKCE, WebAuthn/passkeys, SSO

Lifecycle & integration
Joiner-mover-leaver (JML), provisioning/deprovisioning, SCIM 2.0, REST APIs, webhooks, directory sync

Automation & engineering
Python, PowerShell, Bash, SQL/PostgreSQL, Terraform/IaC, Git, Linux, Docker

Selected IAM Engineering Projects

IAM Lifecycle Automation Engine
Designed a portfolio project for identity provisioning between a source-of-truth system and directory services, with role assignment, dry-run support, structured audit logs, and SCIM 2.0-oriented integration. Focus: Python · SCIM 2.0 · RBAC · audit logging

Enterprise SSO & Federation Lab
Implemented a Keycloak-focused SSO lab exploring OIDC and SAML 2.0 federation, PKCE authorization flows, token handling, and session management. Focus: Keycloak · OIDC · SAML 2.0 · PKCE

Cloud IAM & Governance Analyzer
Developed an IAM analysis concept to evaluate AWS IAM and Entra ID policy configurations, surface excessive permissions and access drift, and support least-privilege reviews and permission-boundary enforcement. Focus: AWS IAM · Microsoft Entra ID · Python · least privilege

Identity Threat Detection & Response
Created detection and response playbook concepts for identity-centric attack patterns, including token replay, MFA fatigue, and privilege escalation, with SIEM-oriented logging and alerting. Focus: SIEM · identity events · detection rules · incident response

Zero-Trust Identity Gateway
Explored reverse-proxy enforcement of identity verification, posture checks, and step-up authentication before application access, with attention to session security. Focus: Zero Trust · OAuth 2.0 · gateway controls · session security

Directory Synchronization Engine
Designed a synchronization utility concept across LDAP, Active Directory, and cloud identity providers, using delta detection and event logging to improve visibility into identity changes. Focus: LDAP · Active Directory · synchronization · event logs

Identity Engineering Approach

- Lifecycle first: connect identity sources to directories and applications; make provisioning and deprovisioning repeatable, reviewable, and auditable.
- Least privilege by design: map roles to required permissions, examine effective access, and use periodic reviews to identify unnecessary entitlements.
- Secure federation: understand trust relationships, token claims, session lifetime, and revocation across OIDC and SAML-based integrations.
- Threat-informed controls: apply offensive-security knowledge to identity risks such as privilege escalation, risky group membership, token abuse, and MFA fatigue.
- Operational discipline: prefer automation with dry-run support, structured logs, version control, and documented validation steps.

Technical Toolkit

Identity & directory
Microsoft Entra ID, AWS IAM, Keycloak, OpenLDAP, Samba, LDAP, Active Directory concepts

Protocols & security
OAuth 2.0, OIDC, SAML 2.0, JWT, PKCE, WebAuthn, MFA, Conditional Access, RBAC/ABAC

Scripting & APIs
Python, PowerShell, Bash, REST APIs, webhooks, SQL, PostgreSQL

Infrastructure & workflow
Linux, Docker, Terraform/IaC, Git, GitHub, security logging, SIEM concepts